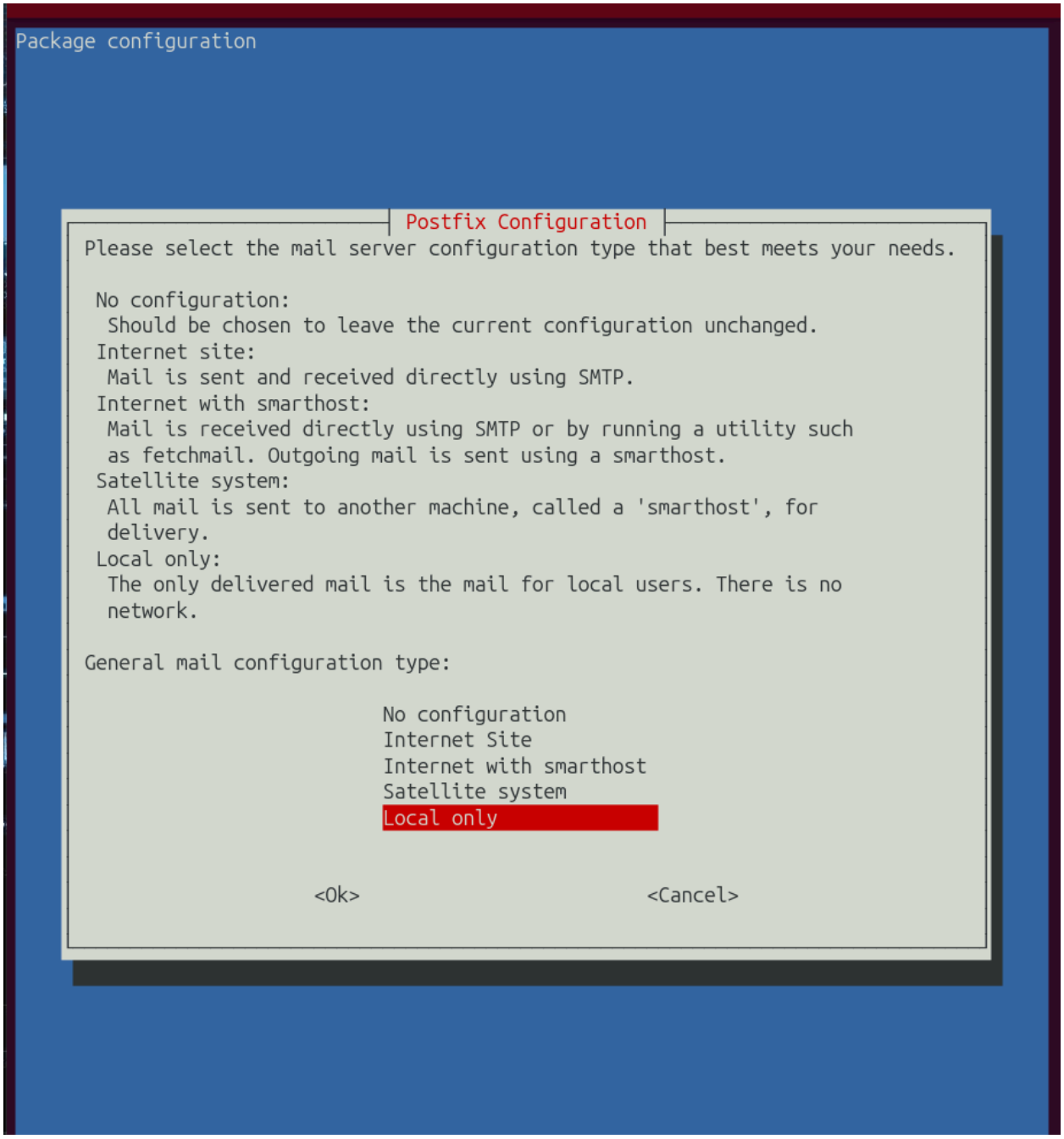
Name: Siddharth Prabhudesai **ID:TSV-919**

1. Configure SMTP server

Configured a local mail server using Postfix to send and receive emails within the same system. The server was set to “Local only”, allowing mail communication between local users using SMTP over `localhost`. The setup was verified by sending a test email and successfully reading it from the user mailbox.

```
sigmoid@sigmoid-ThinkPad-T460:~$ sudo apt install postfix
```



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```
setting mynetworks: 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
clearing mailbox_command
setting mailbox_size_limit: 0
setting recipient_delimiter: +
setting inet_interfaces: loopback-only
setting default_transport: error
setting relay_transport: error
setting inet_protocols: all
/etc/aliases does not exist, creating it.
WARNING: /etc/aliases exists, but does not have a root alias.

Postfix (main.cf) is now set up with a default configuration.  If you need to
make changes, edit /etc/postfix/main.cf (and others) as needed.  To view
Postfix configuration values, see postconf(1).

After modifying main.cf, be sure to run 'systemctl reload postfix'.

Setting up libmailutils9t64:amd64 (1:3.20-2) ...
Setting up mailutils (1:3.20-2) ...
update-alternatives: using /usr/bin/frm.mailutils to provide /usr/bin/frm (frm) in auto mo
de
update-alternatives: using /usr/bin/from.mailutils to provide /usr/bin/from (from) in auto
 mode
update-alternatives: using /usr/bin/messages.mailutils to provide /usr/bin/messages (messa
ges) in auto mode
update-alternatives: using /usr/bin/movemail.mailutils to provide /usr/bin/movemail (movem
ail) in auto mode
update-alternatives: using /usr/bin/readmsg.mailutils to provide /usr/bin/readmsg (readmsg
) in auto mode
update-alternatives: using /usr/bin/dotlock.mailutils to provide /usr/bin/dotlock (dotlock
) in auto mode
update-alternatives: using /usr/bin/mail.mailutils to provide /usr/bin/mailx (mailx) in au
to mode
Setting up libmu-dbm9t64:amd64 (1:3.20-2) ...
Processing triggers for libc-bin (2.42-0ubuntu3) ...
Processing triggers for rsyslog (8.2504.0-1ubuntu2) ...
Processing triggers for ufw (0.36.2-9) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for postfix (3.10.2-1ubuntu2) ...
Restarting postfix
sigmoid@sigmoid-ThinkPad-T460:~$
```

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```
sigmoid@sigmoid-ThinkPad-T460:~$ sudo nano  /etc/postfix/main.cf
[sudo: authenticate] Password:
sigmoid@sigmoid-ThinkPad-T460:~$ sudo systemctl enable postfix
Synchronizing state of postfix.service with SysV service script with /usr/lib/systemd/syst
emd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable postfix
sigmoid@sigmoid-ThinkPad-T460:~$ sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent (main/default instance)
     Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-02 11:12:44 IST; 5h 20min ago
 Invocation: 483557a0d313488283ef34c8a30433fc
       Docs: man:postfix(1)
   Main PID: 2022 (master)
      Tasks: 3 (limit: 8568)
     Memory: 9.1M (peak: 12.6M)
        CPU: 871ms
     CGroup: /system.slice/postfix.service
             ├─ 2022 /usr/lib/postfix/sbin/master -w
             ├─ 2024 qmgr -l -t unix -u
             └─43395 pickup -l -t unix -u -c

Mar 02 11:12:44 sigmoid-ThinkPad-T460 systemd[1]: Starting postfix.service - Postfix Mail>
Mar 02 11:12:44 sigmoid-ThinkPad-T460 postfix[1985]: postfix/postlog: warning: not owned >
Mar 02 11:12:44 sigmoid-ThinkPad-T460 postfix/postfix-script[1985]: warning: not owned by>
Mar 02 11:12:44 sigmoid-ThinkPad-T460 postfix/master[2022]: daemon started -- version 3.1>
Mar 02 11:12:44 sigmoid-ThinkPad-T460 systemd[1]: Started postfix.service - Postfix Mail >
lines 1-19/19 (END)
```

```
No mail for sigmoid
sigmoid@sigmoid-ThinkPad-T460:~$ echo "Test mail body" | mail -s "test Subject" sigmoid@lo
calhost
sigmoid@sigmoid-ThinkPad-T460:~$ mail
"/var/mail/sigmoid": 1 message 1 new
>N   1 sigmoid            Mon Mar  2 16:41  13/475   test Subject
? 1
Return-Path: <sigmoid@sigmoid-ThinkPad-T460>
X-Original-To: sigmoid@localhost
Delivered-To: sigmoid@localhost
Received: by sigmoid-ThinkPad-T460 (Postfix, from userid 1000)
        id E75E8C62C53; Mon, 02 Mar 2026 16:41:48 +0530 (IST)
Subject: test Subject
To: <sigmoid@localhost>
User-Agent: mail (GNU Mailutils 3.20)
Date: Mon,  2 Mar 2026 16:41:48 +0530
Message-Id: <20260302111148.E75E8C62C53@sigmoid-ThinkPad-T460>
From: sigmoid <sigmoid@sigmoid-ThinkPad-T460>

Test mail body
?
```

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2. Non-sudo user creation

Configured a local mail server using Postfix to send and receive emails within the same system. The server was set to “Local only”, allowing mail communication between local users using SMTP over `localhost`. The setup was verified by sending a test email and successfully reading it from the user mailbox.

```
sigmoid@sigmoid-ThinkPad-T460:~$ sudo useradd Sidd
sigmoid@sigmoid-ThinkPad-T460:~$ groups Sidd
Sidd : Sidd
sigmoid@sigmoid-ThinkPad-T460:~$ su - Sidd
Password:
su: Authentication failure
sigmoid@sigmoid-ThinkPad-T460:~$ sudo passwd Sidd
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
sigmoid@sigmoid-ThinkPad-T460:~$ su - Sidd
Password:
su: warning: cannot change directory to /home/Sidd: No such file or directory
$ sudo ls
sudo-rs: I'm sorry Sidd. I'm afraid I can't do that
$ cd /
$ pwd
/
$ sudo ls
sudo-rs: I'm sorry Sidd. I'm afraid I can't do that
$ groups Sidd
Sidd : Sidd
$
```

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3. Global Describe Command

Created a custom shell script named `describe` and placed it in `/usr/local/bin`, which is included in the system PATH. This allows any user to execute the `describe` command from anywhere in the system to list files and directories of the current working directory.

```
#!/bin/bash
ls -1
```

```
sigmoid@sigmoid-ThinkPad-T460:~$ sudo vim /usr/local/bin/describe
sigmoid@sigmoid-ThinkPad-T460:~$ sudo chmod +x /usr/local/bin/describe
sigmoid@sigmoid-ThinkPad-T460:~$ type /usr/local/bin/describe
/usr/local/bin/describe is /usr/local/bin/describe
sigmoid@sigmoid-ThinkPad-T460:~$ cd /etc
sigmoid@sigmoid-ThinkPad-T460:/etc$ describe
ModemManager
NetworkManager
PackageKit
UPower
X11
adduser.conf
aliases
aliases.db
alsa
alternatives
anacrontab
apm
apparmor
apparmor.d
apport
apt
avahi
bash.bashrc
bash_completion
bash_completion.d
bindresvport.blacklist
binfmt.d
bluetooth
```

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4. Find & extract compressed file task

Simulated a scenario where a compressed file named `research` (with any compression extension like `.tar.gz`, etc.) is placed anywhere in the system. Used the `find` command to locate it, the `file` command to determine its compression type, and appropriate extraction tools (`tar`, etc.) to decompress succesfully.

```
sigmoid@sigmoid-ThinkPad-T460:~$ touch research
sigmoid@sigmoid-ThinkPad-T460:~$ echo "This is Siddharth's  Linux Assignment" > rese
ch
sigmoid@sigmoid-ThinkPad-T460:~$ cat research
This is Siddharth's  Linux Assignment
sigmoid@sigmoid-ThinkPad-T460:~$
sigmoid@sigmoid-ThinkPad-T460:~$
sigmoid@sigmoid-ThinkPad-T460:~$ gzip research
sigmoid@sigmoid-ThinkPad-T460:~$ ls
 1                    kernel.txt
 201812301530.45      linux
 Desktop              logs.txt
 Documents            ls
 Downloads            ls.txt
 Music                m.txt
 Pictures             mac2.txt
 Public               mbox
 Templates            my_backup
 Videos               my_host.txt
 a                    my_project
 abc.txt              n
 archive              nmy_backup
 b                    nohup.out
 b.txt               'original-78698365390c7e90bf05e637ba0216a5.png?resize=1024x768'
 backup               output.txt
 c.txt                p.txt
 change.patch         processes.txt
 data.txt             report.txt
 dir1                 research.gz
 dir3                 s_link
 err.txt              script.sh
 etc                  sed
 etc-backup           signal
 etc.tar              signalDemo.cpp
 etc.tar.gz           snap
 exclude_files.txt    softLink
 file.txt             sshd_config
 file4.txt            symlink_p.txt
 first                wget-imgs
 hoops                wget-log
 jakesdir
sigmoid@sigmoid-ThinkPad-T460:~$
```

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```
exclude_files.txt   softLink
file.txt            sshd_config
file4.txt           symlink_p.txt
first               wget-imgs
hoops               wget-log
jakesdir
sigmoid@sigmoid-ThinkPad-T460:~$ tar -czvf research.tar.gz research
research
sigmoid@sigmoid-ThinkPad-T460:~$ ls
1                   kernel.txt
201812301530.45     linux
Desktop             logs.txt
Documents           ls
Downloads           ls.txt
Music               m.txt
Pictures            mac2.txt
Public              mbox
Templates           my_backup
Videos              my_host.txt
a                   my_project
abc.txt             n
archive             nmy_backup
b                   nohup.out
b.txt              'original-78698365390c7e90bf05e637ba0216a5.png?resize=1024x768'
backup              output.txt
c.txt               p.txt
change.patch        processes.txt
data.txt            report.txt
dir1                research
dir3                research.tar.gz
err.txt             s_link
etc                 script.sh
etc-backup          sed
etc.tar             signal
etc.tar.gz          signalDemo.cpp
exclude_files.txt   snap
file.txt            softLink
file4.txt           sshd_config
first               symlink_p.txt
hoops               wget-imgs
jakesdir            wget-log
sigmoid@sigmoid-ThinkPad-T460:~$ rm research
sigmoid@sigmoid-ThinkPad-T460:~$
```

```
sigmoid@sigmoid-ThinkPad-T460:~$ sudo mv research.tar.gz first
sigmoid@sigmoid-ThinkPad-T460:~$ ls first
research.tar.gz  users.txt
sigmoid@sigmoid-ThinkPad-T460:~$
```

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```
sigmoid@sigmoid-ThinkPad-T460:~$ file first/research.tar.gz
first/research.tar.gz: gzip compressed data, from Unix, original size modulo 2^32 1024
0
```

```
sigmoid@sigmoid-ThinkPad-T460:~$ tar -xvf  first/research.tar.gz
research
sigmoid@sigmoid-ThinkPad-T460:~$ ls
1                    kernel.txt
201812301530.45      linux
Desktop              logs.txt
Documents            ls
Downloads            ls.txt
Music                m.txt
Pictures             mac2.txt
Public               mbox
Templates            my_backup
Videos               my_host.txt
a                    my_project
abc.txt              n
archive              nmy_backup
b                    nohup.out
b.txt               'original-78698365390c7e90bf05e637ba0216a5.png?resize=1024x768'
backup               output.txt
c.txt                p.txt
change.patch         processes.txt
data.txt             report.txt
dir1                 research
dir3                 s_link
err.txt              script.sh
etc                  sed
etc-backup           signal
etc.tar              signalDemo.cpp
etc.tar.gz           snap
exclude_files.txt    softLink
file.txt             sshd_config
file4.txt            symlink_p.txt
first                wget-imgs
hoops                wget-log
jakesdir
sigmoid@sigmoid-ThinkPad-T460:~$ cat research
This is Siddharth's  Linux Assignment
sigmoid@sigmoid-ThinkPad-T460:~$
```

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5. File Permission Restriction Without Using chmod

Configured the system using `umask 777` so that any newly created file has no permissions (000). This ensures users cannot read, write, or execute newly created files unless permissions are explicitly modified. The requirement was completed without using the `chmod` command.

```
sigmoid@sigmoid-ThinkPad-T460:~$ sudo nano /etc/profile
[sudo: authenticate] Password:
sigmoid@sigmoid-ThinkPad-T460:~$ source /etc/profile
sigmoid@sigmoid-ThinkPad-T460:~$
sigmoid@sigmoid-ThinkPad-T460:~$ touch testfile
sigmoid@sigmoid-ThinkPad-T460:~$ ls -l testfile
---------- 1 sigmoid sigmoid 0 Mar  2 18:10 testfile
sigmoid@sigmoid-ThinkPad-T460:~$
```

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6. Showtime service

Created a custom systemd service named `showtime` that executes a shell script in the background. Once started, the service writes the current date and time into a file in the user's home directory every minute. The service supports `start`, `stop`, and `status` commands using `systemctl`, demonstrating background process and service management in Linux.

```
sigmoid@sigmoid-ThinkPad-T460:~$ ls -l /usr/local/bin/showtime.sh
-rwxr-xr-x 1 root root 100 Mar  2 19:12 /usr/local/bin/showtime.sh
sigmoid@sigmoid-ThinkPad-T460:~$ sudo nano /etc/systemd/system/showtime.service
sigmoid@sigmoid-ThinkPad-T460:~$ sudo systemctl daemon-reload
sigmoid@sigmoid-ThinkPad-T460:~$ sudo systemctl enable showtime
Created symlink '/etc/systemd/system/multi-user.target.wants/showtime.service' → '/etc/sys
temd/system/showtime.service'.
sigmoid@sigmoid-ThinkPad-T460:~$ sudo systemctl start showtime
sigmoid@sigmoid-ThinkPad-T460:~$ sudo systemctl status showtime
● showtime.service - Show Time Every Minute
     Loaded: loaded (/etc/systemd/system/showtime.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-02 19:13:49 IST; 9s ago
 Invocation: 6fa6b890b5df4fe9b9f40c1c23f5d4a3
   Main PID: 10273 (showtime.sh)
      Tasks: 2 (limit: 8570)
     Memory: 1.3M (peak: 1.7M)
        CPU: 42ms
     CGroup: /system.slice/showtime.service
             ├─10273 /bin/bash /usr/local/bin/showtime.sh
             └─10276 sleep 60

Mar 02 19:13:49 sigmoid-ThinkPad-T460 systemd[1]: Started showtime.service - Show Time Ev>
sigmoid@sigmoid-ThinkPad-T460:~$ cat /home/sigmoid/showtime.txt
Mon Mar  2 19:13:49 IST 2026
sigmoid@sigmoid-ThinkPad-T460:~$
```

```
  GNU nano 8.4                    /etc/systemd/system/sho
[Unit]
Description=Show Time Every Minute

[Service]
ExecStart=/usr/local/bin/showtime.sh
User=sigmoid
Restart=always

[Install]
WantedBy=multi-user.target
```

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```
sigmoid@sigmoid-ThinkPad-T460:~$ cat /home/sigmoid/showtime.txt
Mon Mar  2 19:13:49 IST 2026
Mon Mar  2 19:14:49 IST 2026
Mon Mar  2 19:15:49 IST 2026
Mon Mar  2 19:16:49 IST 2026
Mon Mar  2 19:17:49 IST 2026
Mon Mar  2 19:18:49 IST 2026
Mon Mar  2 19:19:49 IST 2026
Mon Mar  2 19:20:49 IST 2026
Mon Mar  2 19:21:49 IST 2026
Mon Mar  2 19:22:49 IST 2026
Mon Mar  2 19:23:49 IST 2026
Mon Mar  2 19:24:49 IST 2026
Mon Mar  2 19:25:49 IST 2026
Mon Mar  2 19:26:49 IST 2026
Mon Mar  2 19:27:49 IST 2026
Mon Mar  2 19:28:49 IST 2026
Mon Mar  2 19:29:49 IST 2026
Mon Mar  2 19:30:49 IST 2026
Mon Mar  2 19:31:49 IST 2026
Mon Mar  2 19:32:49 IST 2026
Mon Mar  2 19:33:49 IST 2026
sigmoid@sigmoid-ThinkPad-T460:~$ sudo systemctl status showtime
```